

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Solution Requirements

ShopEZ E-Commerce Application

Phase: Requirement Analysis Phase

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Solution Requirements

1. Introduction

This document defines the **Functional** and **Non-Functional Requirements** for the **ShopeZ E-Commerce Application**. The system enables customers to browse products, manage carts, place orders, make secure payments, and track deliveries, while administrators handle product listings, inventory, and order management. Each requirement is uniquely identified for traceability across analysis, design, implementation, testing, and deployment.

2 Functional Requirements

FR-01 to FR-05 — User Management

Req. ID	Requirement Description	Priority	USN
FR-01	The system shall allow new customers to register using name, email, and password.	Must Have	USN-1
FR-02	The system shall authenticate registered users using email and password credentials	Must Have	USN-2
FR-03	The system shall securely store passwords using encryption techniques.	Must Have	USN-2
FR-04	The system shall provide role-based access control for Customers and Administrators.	Must Have	USN-7
FR-05	The system shall allow users to update their profile information.	Should Have	USN-3

FR-06 to FR-10 — ProductCatalogue & ShoppingCart

Req. ID	Requirement Description	Priority	USN
FR-06	The system shall allow customers to browse products by category and search keywords.	Must Have	USN-3
FR-07	The system shall store product details and inventory in the database.	Must Have	USN-8
FR-08	The system shall allow customers to add products to the cart and update quantities.	Must Have	USN-4
FR-09	The system shall provide checkout functionality with address and payment options.	Must Have	USN-5
FR-10	Administrators shall be able to view and manage product listings.	Must Have	USN-8

FR-11 to FR-15 — Shopping Cart & Checkout

Req. ID	Requirement Description	Priority	USN
FR-11	Administrators shall update order status (Processing, Shipped, Delivered)..	Must Have	USN-9
FR-12	The system shall notify customers when order status changes..	Must Have	USN-9
FR-13	Customers shall be able to provide feedback after order delivery.	Must Have	USN-10
FR-14	Administrators shall view customer feedback and ratings.	Should Have	USN-10
FR-15	The system shall generate sales reports and analytics..	Should Have	USN-10

3. Non-Functional Requirements

Req. ID	Category	Requirement Description	Priority
NFR-01	Security	User passwords must be encrypted using bcrypt before storage.	Must Have
NFR-02	Security	Protected APIs shall require authentication tokens.	Must Have
NFR-03	Security	Only administrators can modify product or order data.	Must Have
NFR-04	Performance	Product and order retrieval should complete within 2 seconds.	Should Have
NFR-05	Reliability	System shall maintain data integrity for orders and payments..	Must Have
NFR-06	Usability	User interface shall be responsive across desktop and mobile devices.	Must Have
NFR-07	Availability	System shall be available 99% of the time.	Should Have

Req. ID	Category	Requirement Description	Priority
NFR-08	Maintainability	Backend shall follow MVC architecture.	Should Have
NFR-09	Scalability	System shall support increasing numbers of users and orders.	Could Have

4. Requirements Traceability Matrix

Req. ID	Design Artefact	Implementation File	UAT Test Case
FR-01, FR-02,	User Architecture Design	userController.js	UAT-001
FR-03, FR-04	Security&Access Control Design	authMiddleware.js	UAT-002
FR-06, FR-07	Product Catalogue Module	productController.js	UAT-003
FR-08, FR-09	Cart & Checkout Module	cartController.js	UAT-004
FR-10, FR-11	AdminOrder Mngement	adminController.js	UAT-005
FR-13, FR-14	Feedback Management Module	feedbackController.js	UAT-006